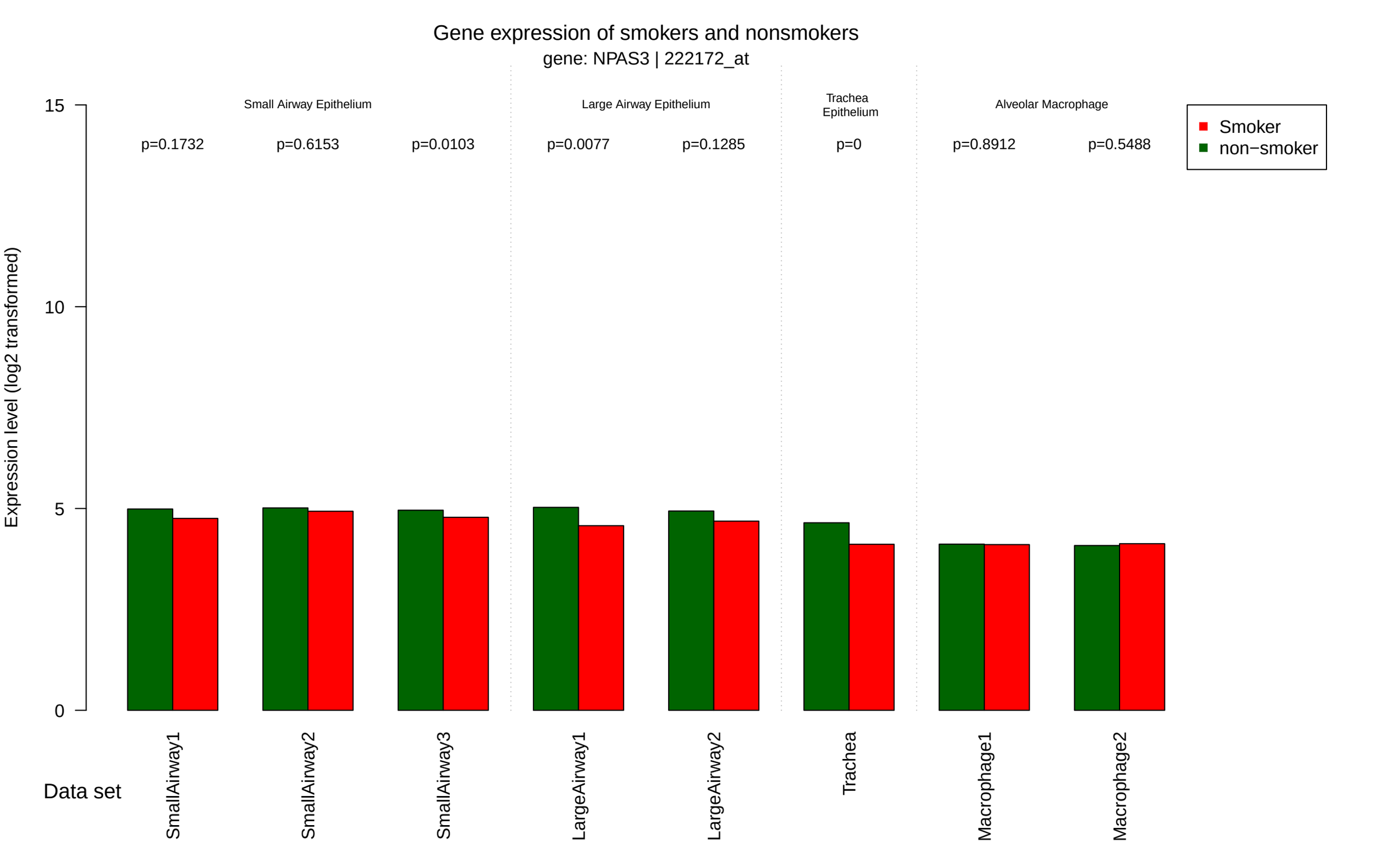
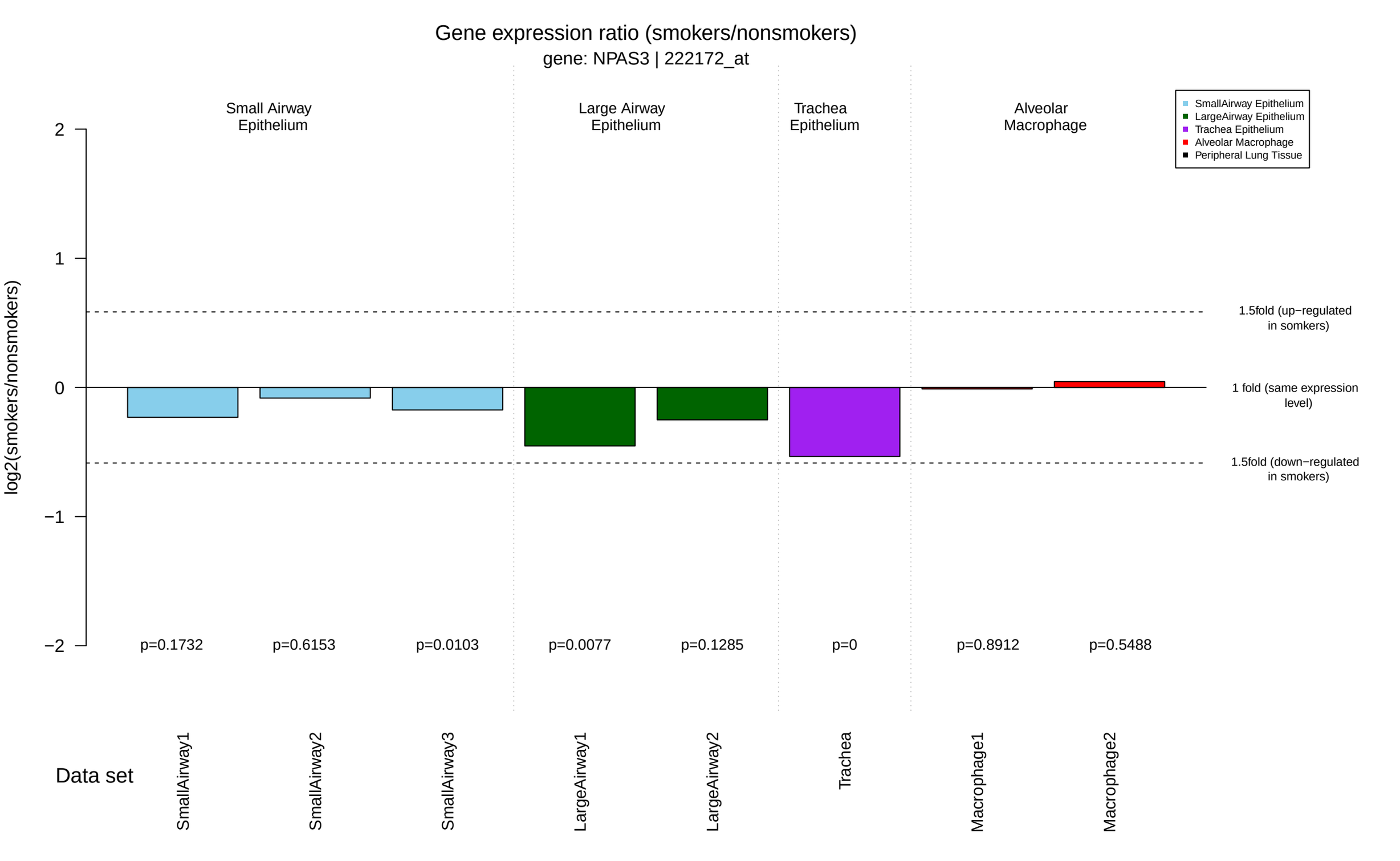
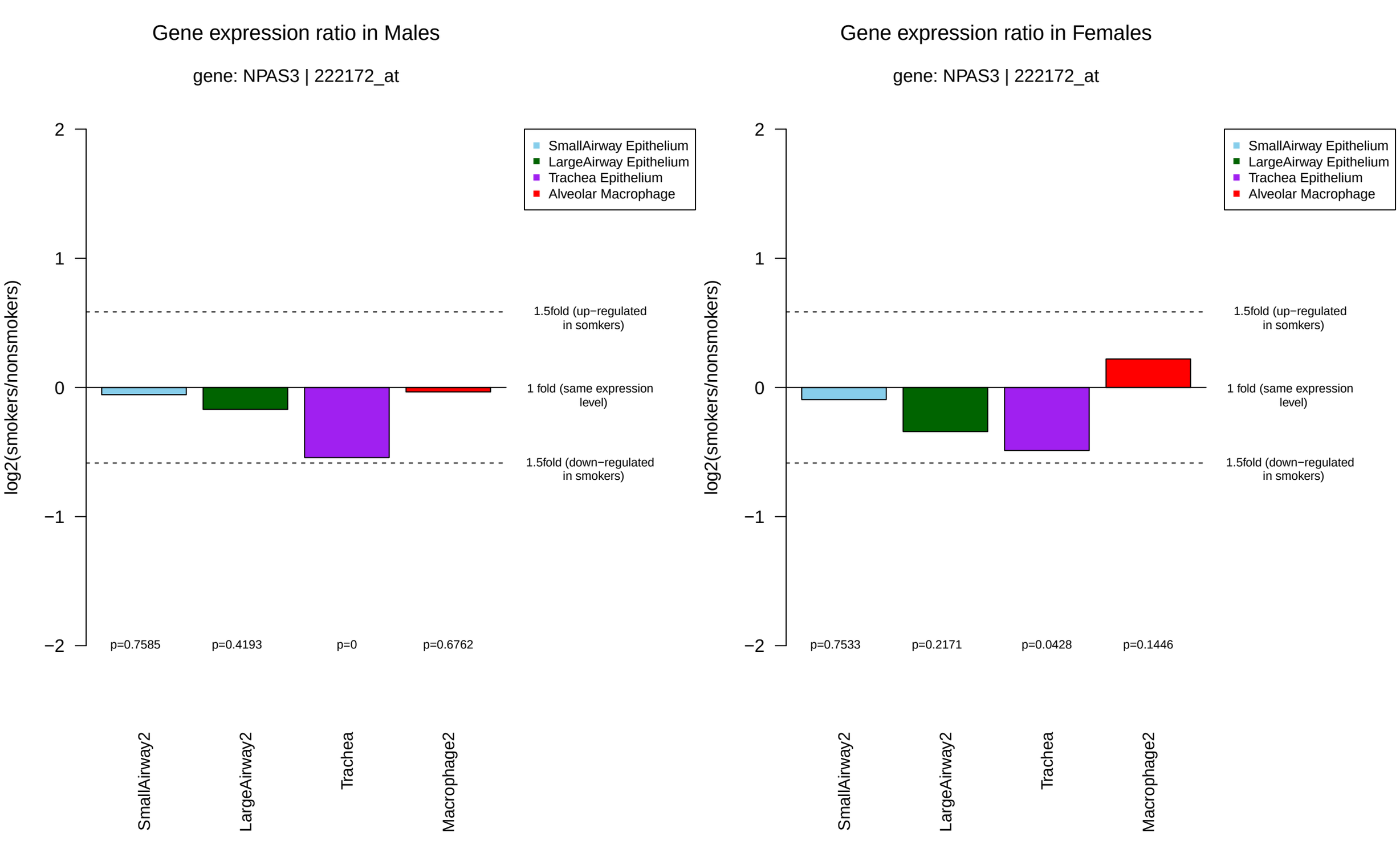
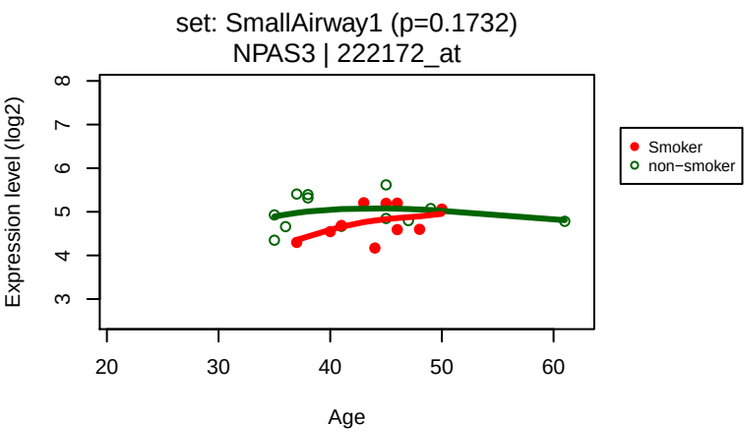
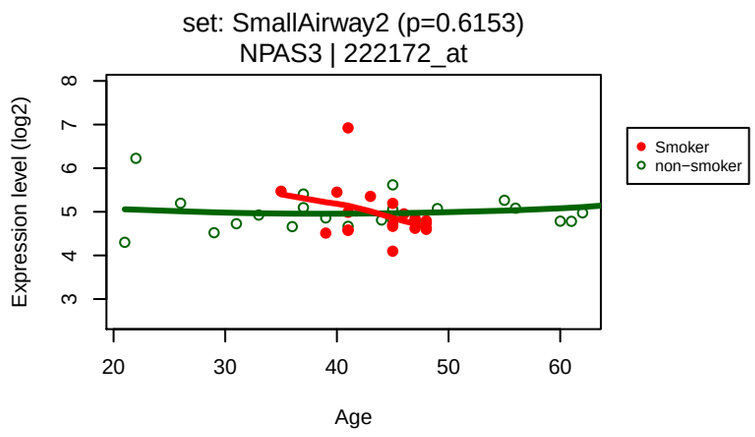
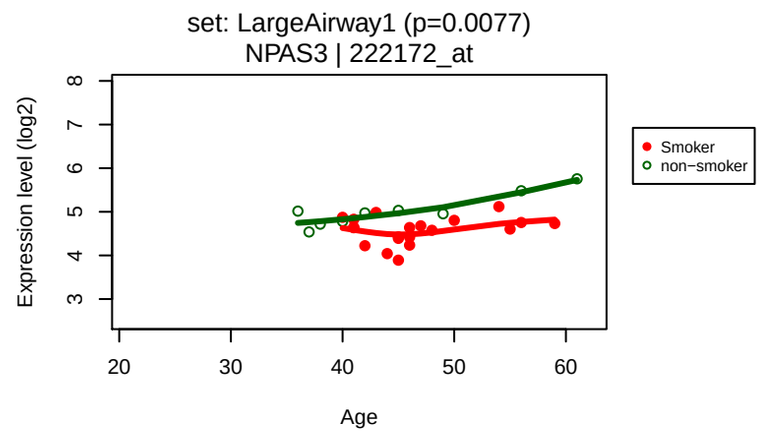

Gene expression of smokers and nonsmokers
gene: NPAS3 | 222172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1732
p=0.6153
p=0.0103
p=0.0077
p=0.1285
p=0
p=0.8912
p=0.5488
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

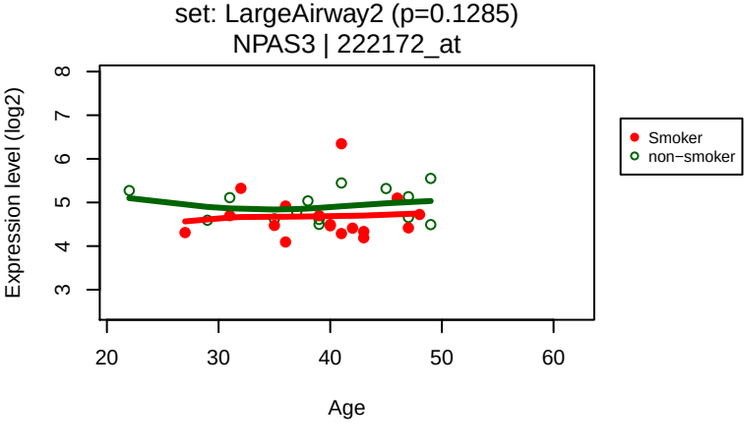

Gene expression ratio (smokers/nonsmokers)
gene: NPAS3 | 222172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1732
p=0.6153
p=0.0103
p=0.0077
p=0.1285
p=0
p=0.8912
p=0.5488
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

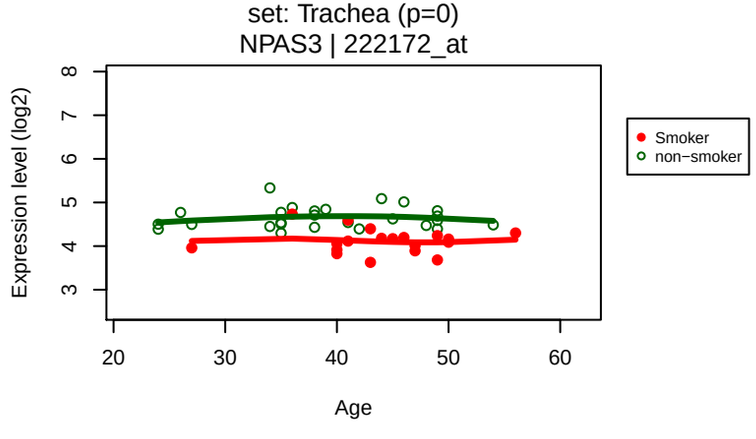

Gene expression ratio in Males
gene: NPAS3 | 222172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7585
p=0.4193
p=0
p=0.6762
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS3 | 222172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7533
p=0.2171
p=0.0428
p=0.1446
SmallAirway2
LargeAirway2
Trachea
Macrophage2

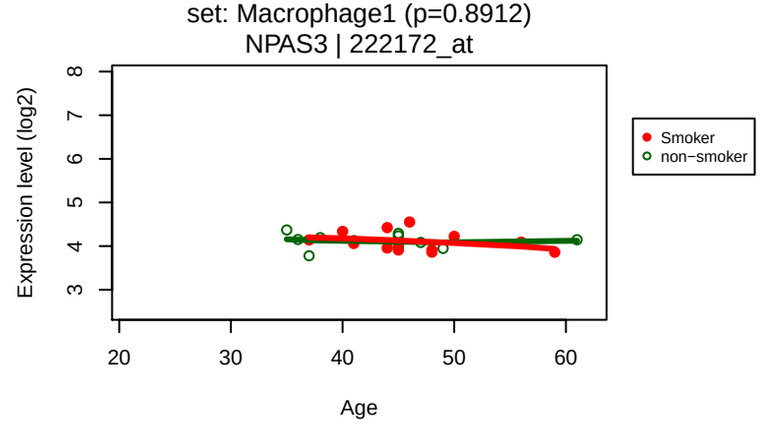

set: SmallAirway1 (p=0.1732)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker

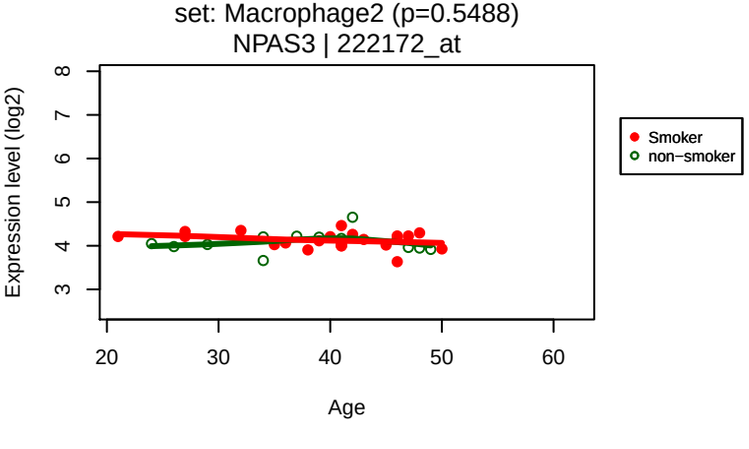

set: SmallAirway2 (p=0.6153)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker

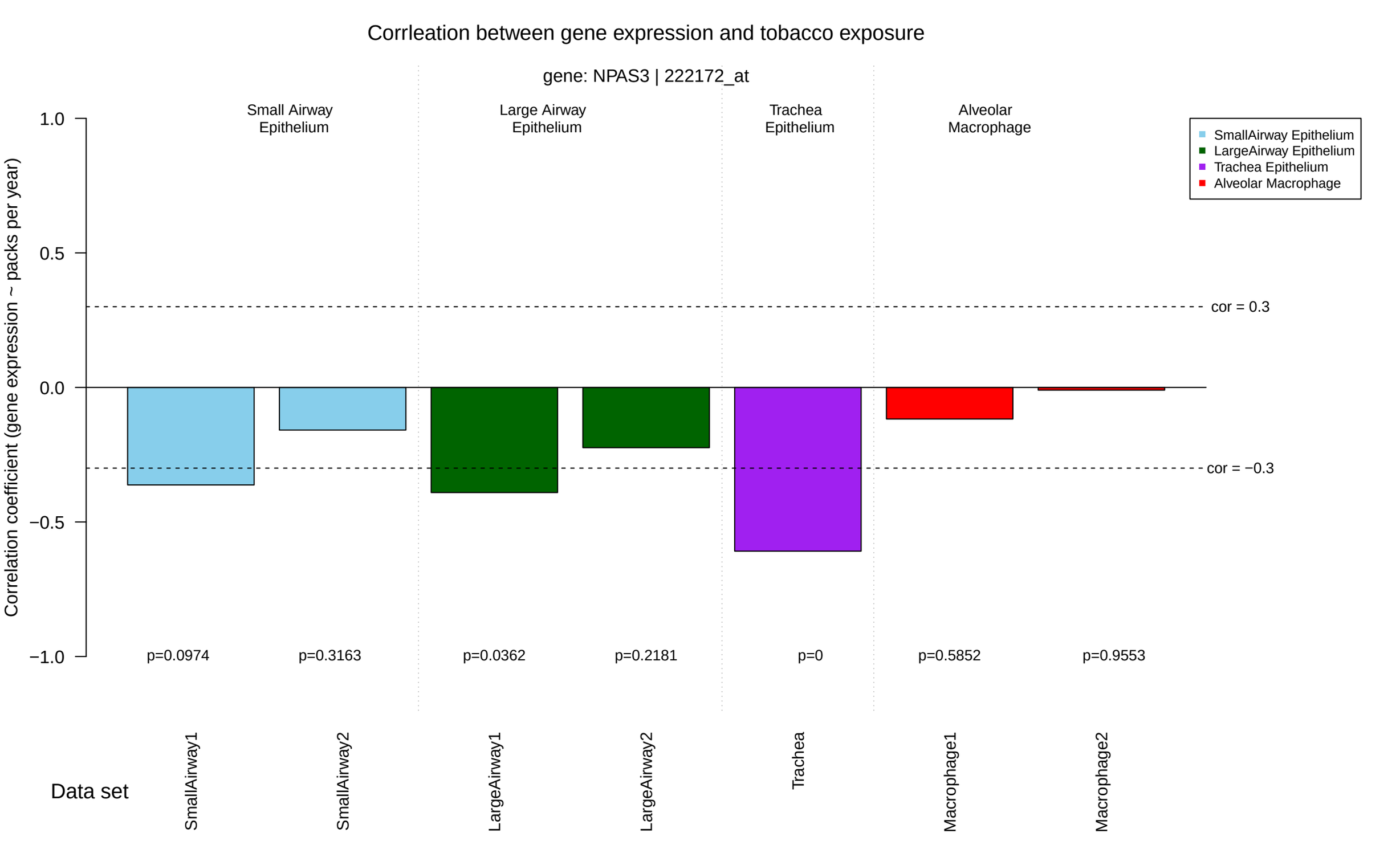

set: LargeAirway1 (p=0.0077)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1285)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8912)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5488)
NPAS3 | 222172_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS3 | 222172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0974
p=0.3163
p=0.0362
p=0.2181
p=0
p=0.5852
p=0.9553
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2